
UNIVERSITY OF CAPE TOWN
Test - 21 November 2022
Investment Decision Theory and Application
ECO4102Z/ECO4122Z

Time: 2 $\frac{1}{2}$ hours

Maximum Marks: 75

INSTRUCTIONS TO CANDIDATES

1. You have 5 minutes at the start of the examination in which to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made on this examination paper. You then have three hours to complete the paper.
2. You must not start writing your answers in the booklet until instructed to do so by the invigilator.
3. Mark allocations are shown in brackets [].
4. Attempt all 6 questions, beginning your answer to each full question on a new page.
5. Show all calculations where this is appropriate.
6. Write legibly using blue or black ink.

1. Godfrey Ndlovu, is reviewing the defined benefit pension plan of the University of Cape Town(UCT). Next month, UCT intends to implement a new Human Resources and Talent Management strategic plan, in-line with the university's Vision 2030. Part of this involves a review of the retirement age for full benefits under the plan from age 65 to age 60. The median age of UCT's workforce is 47 years. Godfrey is responsible for the pension plan's investment policy and strategic asset allocation decisions. The goals of the plan include achieving a minimum expected return of 7.6% with expected standard deviation no greater than 15.0%. Godfrey is evaluating the current asset allocation (Table 1) and selected financial information for the UCT plan (Table 2). There is an ongoing debate within UCT about the pension plan's investment policy statement (IPS). Extracts from the two investment policy statements under consideration are shown in Table 3

Table 1: **Current Asset Allocation**

Asset Class	Proportion(%)
International Equities (MSCI World, excluding South Africa)	9
South African bonds	40
JSE small capitalization equities	16
JSE large capitalization equities	30
Cash	5

Table 2: **UCT Selected Financial Information (in millions of Rand)**

UCT total assets	16,000
<i>Pension plan data:</i>	
Plan assets	6,040
Plan liabilities	9,850

- (a) Determine, for each of the following components, whether IPS A or IPS B (see Table 3) is the most appropriate for the pension plan of UCT. Justify each response with two reasons.
- Return Requirement.
 - Risk Tolerance.
 - Time horizon
 - Liquidity requirements

Note: Some components of IPS A may be appropriate, while other components of IPS B may be appropriate.

[12 marks]

- (b) To assist Godfrey, UCT has hired two pension investment specialists, Giacomo Fiore and Tylin Naicker. Giacomo believes that the pension fund must be invested

Table 3: **Investment Policy Statements**

	IPS A	IPS B
Return requirement	Plan's objective is to outperform the relevant benchmark return by a substantial margin.	Plan's objective is to match the relevant benchmark return.
Risk tolerance	Plan has a high risk tolerance because of the long-term nature of the plan and its liabilities.	Plan has a low risk tolerance because of its limited ability to assume substantial risk
Time horizon	Plan has a very long time horizon because of the plan's infinite life.	Plan has a shorter time horizon than in the past because of plan demographics.
Liquidity	Plan needs moderate level of liquidity to fund monthly benefit payments.	Plan has minimal liquidity needs.

Table 4: **Recommended Asset allocations (in %)**

	Current	Giacomo	Tylin
JSE large capitalization equities	30%	22%	40%
JSE small capitalization equities	16	8	22
International equities (MSCI World ex-South Africa)	9	9	16
South African bonds	40	50	17
Cash	5	11	5
Total	100%	100%	100%
Expected portfolio return	8.2%	7.2%	10.1%
Expected portfolio volatility (standard deviation)	15.2%	13.8%	20.2%

to reflect a low risk tolerance, whereas Tylin believes the pension fund must be invested to achieve the highest possible returns. The fund's current asset allocation and the allocations recommended by Giacomo and Tylin are shown in Table 4. Select which of the three asset allocations in Table 4 is most appropriate for UCT's pension plan. Explain how your selection meets each of the following objectives or constraints for the plan:

- i. Return requirement
- iii. Risk tolerance
- iiii. Liquidity requirements

[9 marks]

[Total: 21 marks]**SUGGESTED SOLUTION**

a Any of the following

- i. Return Requirement: IPS B has the appropriate language/terminology. Since the Plan is currently under-funded, the primary objective should be to make the pension fund financially stronger. The risk inherent in attempting to maximize total returns would be inappropriate.
- ii. Risk Tolerance: IPS B has the appropriate language. Because of the fund's underfunded status, the Plan has limited risk tolerance; should the fund incur a substantial loss, payments to beneficiaries could be jeopardized.
- iii. Time Horizon: IPS B has the appropriate language. Although going-concern pension plans usually have a long time horizon, the UCT plan has a shorter time horizon because of the reduced retirement age and the relatively high median age of the workforce.
- iv. Liquidity: IPS A has the appropriate language. Because of the early retirement feature starting next month and the age of the work force (which indicates an increasing number of retirees in the near future), the Plan needs a moderate level of liquidity in order to fund monthly benefit payments.

b The current portfolio is the most appropriate choice for the pension plan's asset allocation. The current portfolio offers: i. An expected return that exceeds the Plan's return requirement; ii. An expected standard deviation that only slightly exceeds the Plan's target; and, iii. A level of liquidity that should be sufficient for future needs. The higher expected return will ameliorate the Plan's under-funded status somewhat, and the change in the fund's risk profile will be minimal. The portfolio has significant allocations to RSA bonds (40 percent) and large-cap equities (30 percent) in addition to cash (5 percent). The availability of these highly liquid assets should be sufficient to fund monthly benefit payments when the early retirement feature takes effect next month, particularly in view of the stable income flows from these investments.

The Giacomo portfolio offers:

- i. An expected return that is slightly below the Plan's requirement;
- ii. An expected standard deviation that is substantially below the Plan's target; and,
- iii. A level of liquidity that should be more than sufficient for future needs.

Given the Plan's under-funded status, the portfolio's level of risk is unacceptable. The Tylin portfolio offers:

- i. An expected return that is substantially above the Plan's requirement;
- ii. An expected standard deviation that far exceeds the Plan's target; and
- iii. A level of liquidity that should be sufficient for future needs. Given the Plan's under-funded status, the portfolio's level of risk is unacceptable.

2. An investor makes decisions based on the utility function $U(w) = w - 6w^2$, where w is the investor's wealth in millions of rand (Rm).

- (a) Demonstrate that the investor has both increasing absolute and relative risk aversion. Show all calculations. [4 marks]

The investor has R50,000 to invest over a 1-year period and has no other wealth. They have three options:

- i. Invest in a risk-free account. There will be no change in the value of the investment over 1 year.
- ii. Invest in an asset that will give a 60% return over 1 year with probability 0.2, a 20% return with probability 0.7 and a -40% return with probability 0.1.
- iii. Invest in an asset that will give a 30% return with probability 0.5 and a 20% return with probability 0.5.

The investor makes no allowance for discounting when making investment decisions. The investor must invest the whole R50,000 in a single option.

- (b) Determine which option the investor should choose to maximise their expected utility at the end of the year. [6 marks]
- (c) Comment on why the investor could not use $U(w)$ to choose from the above options if their initial wealth was R65,000. [3 marks]

[Total: 13 marks]

SUGGESTED SOLUTION

a.

$$U(w) = w - 6w^2$$

$$U'(w) = 1 - 12w[1/2]$$

$$U''(w) = -12[1/2]$$

$$A(w) = -U''(w)/U'(w) = 12/(1 - 12w)[1/2]$$

$$A'(w) = 144(1 - 2w)^2 > 0[1/2]$$

$$R(w) = w \times -U''(w)/U'(w) = 12w/(1 - 12w)[1/2]$$

$$R'(w) = 12/(1 - 12w) + 144w/(1 - 12w)^2 = 12/(1 - 12w)^2 > 0[1/2]$$

Option A:

$$w = 0.05[1]$$

$$E(U(w)) = 0.05 - 6 \times 0.05^2 = 0.035[1]$$

Option B:

$w = 0.08$ with probability 0.2, $w = 0.06$ with probability 0.7, $w = 0.03$ with probability 0.1[1]

$$\begin{aligned} E(U(w)) &= 0.2 \times (0.08 - 6 \times 0.08^2) + 0.7 \times (0.06 - 6 \times 0.06^2) + 0.1 \times (0.03 - 6 \times 0.03^2) \\ &= 0.03766[1] \end{aligned}$$

Option C:

$w = 0.065$ with probability 0.5, $w = 0.06$ with probability 0.5 [1]

$$\begin{aligned} E(U(w)) &= 0.5 \times (0.065 - 6 \times 0.0652) + 0.5 \times (0.06 - 6 \times 0.062) \\ &= 0.039025[1] \end{aligned}$$

The investor is likely to choose option C as it gives the highest expected utility. [1]

- b. One of the assumptions of the expected utility theorem is non-satiation. [1]. In terms of utility functions this means $U'(w) > 0$. [1] For this investor $U'(w) = 1 - 12w$. If $1 - 12w > 0$ then $w < 0.08333$, *i.e.* R83,333. [1]. If the investor had R65,000 then his wealth would exceed R83,333 under any outcome with a return over 28% (e.g. in option 2 if the return was 60% or in option 3 if the return was 30%). [1] Therefore this utility function could not be used to assess the preferred option. [1] [Marks available 5, maximum 3]

3. Define the following terms:

- (a) Mean-preserving spread [1 mark]
 (b) Second-order stochastic dominance [1 mark]

Consider the following investments whose expected returns(%), have the following probability distribution(s):

Return	Probability	
	Investment 1	Investment 2
6%	0.25	0.00
7%	0.25	0.75
8%	0.25	0.00
9%	0.25	0.25

- (c) Compare the two investments using the second-order stochastic dominance criterion. [5 marks]
 (d) Explain the practical limitation(s) of the concept of stochastic dominance in modelling investor choice. [1 mark]

[Total: 8 marks]

- a. a gamble X is said to be a mean preserving spread of Y if the two have the same expectation and Y second-order stochastically dominates X- some students may use a graph to illustrate
 b. 2 SOSD 1- students to show calculations
 c. It is a partial ordering, and may fail in certain instances as in the example above[1]; there is no acceptable way to use data to test for stochastic dominance. [1]

4. Suppose that there are two risky assets: X and Y, with the following properties.

	Expected Return	Standard Deviation
Asset A	0.07	0.2
Asset B	0.12	0.4

The risk-free rate is 4 percent, and the correlation between assets X and Y is 0.25.

- (a) Determine the expected return and standard deviation of a portfolio of X and Y that maximizes the Sharpe ratio (i.e. optimal portfolio). Assume no restrictions to short selling. [10 marks]
- (b) Investors can hold any portfolio consisting of asset X and asset Y and also the risk-free asset. In addition, suppose Old Mutual asset management offers a mutual fund with an expected return of 6% and a standard deviation of 10%. All investors have standard mean-variance preferences. Should any investor invest in the Old Mutual's fund? Explain clearly why or why not. [3 marks]

[Total: 13 marks]

Covariance= $0.25 \cdot .2 \cdot .4 = 0.02$ - 1 mark

$V = \begin{pmatrix} 0.04 & 0.02 \\ 0.02 & 0.16 \end{pmatrix}$ 1 mark

$V^{-1} = \begin{pmatrix} 26.67 & -3.33 \\ -3.33 & 26.67 \end{pmatrix}$ - 2.5 marks

$\omega_x = 0.55$ $\omega_y = 0.45$ 2.5 marks

$E(R_p) = 0.0924$; 1 mark

$\sigma_p^2 = 0.05422$ - 1 mark

$\sigma_p = 0.2329$ - 1 mark

5. Consider the following assets in a world where the Capital Asset Pricing Model (CAPM) holds. There are three risky assets and one risk-free asset. No other assets exist in the market.

Asset	Expected return (% p.a.)	Total value of assets in market	Beta
Risky asset A	7.0	10,000,000	β
Risky asset B	13.5	50,000,000	1.5
Risky asset C	x	20,000,000	2.0
Risk-free asset	3.0	40,000,000	n/a

- (a) Determine the expected return on the market portfolio. [2 marks]
- (b) Calculate x . and β . [1; 1 mark]
- (c) Discuss the limitations of the CAPM. [4 marks]

[Total: 8 marks]

$$\left(\frac{13.5\% - 3\%}{EM - 3\%} \right) = 1.5 \quad [1]$$

$$\Rightarrow RM = 10\% \quad [1]$$

$$x - 3\% = 2 \cdot (10\% - 3\%) \quad [1]$$

$$= 17\%$$

$$7\% - 3\% = \text{Beta} \times (12.25\% - 3\%) \quad [1] \Rightarrow \text{beta} = 0.571 \quad [1]$$

may include the following:
There are basic problems in testing the model since, in theory, account has to be taken of the entire investment universe open to investors, not just capital markets [1]

An important asset of most investors, for example, is their human capital (i.e. the value of their future earnings). Models have been developed which allow for decisions over multiple periods and for the optimisation of consumption over time to take account of this [1]

Other versions of the basic CAPM have been produced which allow for taxes, inflation, and also for a situation where there is no riskless asset [1]

In the international situation there is no asset which is riskless for all investors (due to currency risks) so a model has been developed which allows for groups of investors in different countries, each of which considers their domestic currency to be risk-free [1]

There is a discrepancy between the values obtained for the expected return on the market using the security market line and the weightings by market capitalisation [1]

Some other assumptions of CAPM are unrealistic, e.g. everyone has the same estimates for the means/variances/covariances, everyone has the same single time horizon [1]

[Marks available 6, maximum 3]

6. The following information describes an analyst's expected return and risk relationship for the stocks of *A* and *B*.

	Expected Return	Standard Deviation	Beta
Stock A	12.0%	20%	1.3
Stock B	9.0	15	0.7
Market Index	10.0	12	1.0
Risk-free	5		

Using only the data shown in the preceding table:

- (a) Draw and label a graph showing the security market line (SML) and position stocks *A* and *B* relative to it. [5 marks]
- (b) Compute the alphas both for Stock *A* and for Stock *B*. Show your work. [4 marks]

$$R_i - R_f = \alpha + \beta(R_m - R_f)$$

$$\alpha_A = 12\% - [5\% + 1.3\%(10\% - 5\%)] = 0.5\%$$

$$\alpha_B = 9\% - [5\% + 0.7\%(10\% - 5\%)] = 0.5\%$$

In this instance, the alphas are equal and both are positive, so one does not dominate the other. (I had given more marks here assuming you will all get this correct).

- (c) Assume that the risk-free rate increases to 7 percent with the other data in the preceding table remaining unchanged. Explain(in words) the impact of this on the SML and identify the stock providing the higher expected risk-adjusted return and justify your selection. Show your calculations. [3 marks]

By increasing the risk-free rate from 5 percent to 7 percent and leaving all other factors unchanged, the slope of the SML flattens and the expected return per unit of incremental risk becomes less. Using the formula for alpha, the alpha of Stock A increases to 1.1 percent and the alpha of Stock B falls to -0.1 percent. In this situation, the expected return (12.0 percent) of Stock A exceeds its required return (10.9 percent) based on the CAPM. Therefore, Stock A's alpha (1.1 percent) is positive. For StockB, its expected return (9.0 percent) is below its required return (9.1 percent) based on the CAPM. Therefore, Stock B's alpha (-0.1 percent) is negative. Stock A is preferable to Stock B under these circumstances. Calculations of revised alphas:

$$\begin{aligned} \text{Stock A} &= 12\% - [7\% + 1.3(10\% - 7\%)] \\ &= 12\% - 10.95\% = 1.1\% \\ \text{Stock B} &= 9\% - [7\% + 0.7(10\% - 7\%)] \\ &= 9\% - 9.1\% = -0.1\% \end{aligned}$$

[Total: 12 marks]
